Hospital Patient Analysis

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10-09-2026

Introduction

This Tiny Project analyzes hospital patient data using a sample dataset containing patient names, departments, ages, hospital stay, treatment cost, and satisfaction scores.

1. Data Importing

```
patient_data <- read.csv("hospital_patient_data.csv")
patient_data
```

```
##    Patient_Name       Department Age Stay_Days Treatment_Cost
## 1    Aarav Shah       Cardiology  45         5          45000
## 2    Diya Patel General Medicine  32         3          18000
## 3   Rohan Mehta      Orthopedics  58         7          62000
## 4  Ananya Joshi       Pediatrics  12         2          12000
## 5   Kabir Verma       Cardiology  67         6          78000
## 6    Ishita Rao      Dermatology  29         2          15000
## 7   Arjun Desai General Medicine  41         4          28000
## 8    Meera Nair      Orthopedics  52         8             NA
## 9  Vivaan Singh       Pediatrics   8         3          14000
## 10  Riya Kapoor       Cardiology  73         9          92000
## 11  Aditya Jain      Dermatology  35         2          19000
## 12   Kavya Iyer General Medicine  48         5          36000
## 13   Yash Gupta      Orthopedics  61         6          48000
## 14  Neha Sharma       Cardiology  54         7          68000
## 15    Dev Kumar       Pediatrics  15         3          16000
## 16   Pooja Shah      Dermatology  39         4          24000
## 17  Rahul Patel General Medicine  46         4          30000
## 18  Sneha Mehta      Orthopedics  57         7          59000
## 19  Kunal Verma       Cardiology  69         8          85000
## 20    Tanya Rao       Pediatrics  10         2          13000
##    Satisfaction_Score
## 1                 4.5
## 2                 4.2
## 3                 4.0
## 4                 4.7
## 5                 3.8
## 6                 4.4
## 7                 4.3
## 8                 4.1
## 9                 4.8
## 10                3.7
## 11                4.5
## 12                4.2
## 13                4.0
## 14                3.9
## 15                 NA
## 16                4.3
## 17                4.4
## 18                4.0
## 19                3.8
## 20                4.7
```

```
head(patient_data)
```

```
##   Patient_Name       Department Age Stay_Days Treatment_Cost Satisfaction_Score
## 1   Aarav Shah       Cardiology  45         5          45000                 4.5
## 2   Diya Patel General Medicine  32         3          18000                 4.2
## 3  Rohan Mehta      Orthopedics  58         7          62000                 4.0
## 4 Ananya Joshi       Pediatrics  12         2          12000                 4.7
## 5  Kabir Verma       Cardiology  67         6          78000                 3.8
## 6   Ishita Rao      Dermatology  29         2          15000                 4.4
```

```
summary(patient_data)
```

```
##     Patient_Name     Department      Age          Stay_Days    Treatment_Cost
##  Length   :20    Length   :20   Min.   : 8.00   Min.   :2.00   Min.   :12000
##  N.unique :20    N.unique : 5   1st Qu.:31.25   1st Qu.:3.00   1st Qu.:17000
##  N.blank  : 0    N.blank  : 0   Median :45.50   Median :4.50   Median :30000
##  Min.nchar: 9    Min.nchar:10   Mean   :42.55   Mean   :4.85   Mean   :40105
##  Max.nchar:12    Max.nchar:16   3rd Qu.:57.25   3rd Qu.:7.00   3rd Qu.:60500
##                                 Max.   :73.00   Max.   :9.00   Max.   :92000
##                                                               NAs    :1
##  Satisfaction_Score
##  Min.   :3.700
##  1st Qu.:4.000
##  Median :4.200
##  Mean   :4.226
##  3rd Qu.:4.450
##  Max.   :4.800
##  NAs    :1
```

```
str(patient_data)
```

```
## 'data.frame':    20 obs. of  6 variables:
##  $ Patient_Name      : chr  "Aarav Shah" "Diya Patel" "Rohan Mehta" "Ananya Joshi"
...
##  $ Department        : chr  "Cardiology" "General Medicine" "Orthopedics" "Pediatri
cs" ...
##  $ Age               : int  45 32 58 12 67 29 41 52 8 73 ...
##  $ Stay_Days         : int  5 3 7 2 6 2 4 8 3 9 ...
##  $ Treatment_Cost    : num  45000 18000 62000 12000 78000 15000 28000 NA 14000 9200
0 ...
##  $ Satisfaction_Score: num  4.5 4.2 4 4.7 3.8 4.4 4.3 4.1 4.8 3.7 ...
```

2. Data Preprocessing and Cleaning

Checking Missing Values

```
colSums(is.na(patient_data))
```

```
##       Patient_Name         Department                Age          Stay_Days 
##                  0                  0                  0                  0 
##     Treatment_Cost Satisfaction_Score 
##                  1                  1
```

Handling Missing Values

```
patient_data$Treatment_Cost[is.na(patient_data$Treatment_Cost)] <- median(patient_data
$Treatment_Cost, na.rm=TRUE)
patient_data$Satisfaction_Score[is.na(patient_data$Satisfaction_Score)] <- median(pati
ent_data$Satisfaction_Score, na.rm=TRUE)
```

3. Statistical Analysis

```
mean(patient_data$Treatment_Cost)
```

```
## [1] 39600
```

```
median(patient_data$Treatment_Cost)
```

```
## [1] 30000
```

```
var(patient_data$Treatment_Cost)
```

```
## [1] 674673684
```

```
sd(patient_data$Treatment_Cost)
```

```
## [1] 25974.48
```

```
min(patient_data$Treatment_Cost)
```

```
## [1] 12000
```

```
max(patient_data$Treatment_Cost)
```

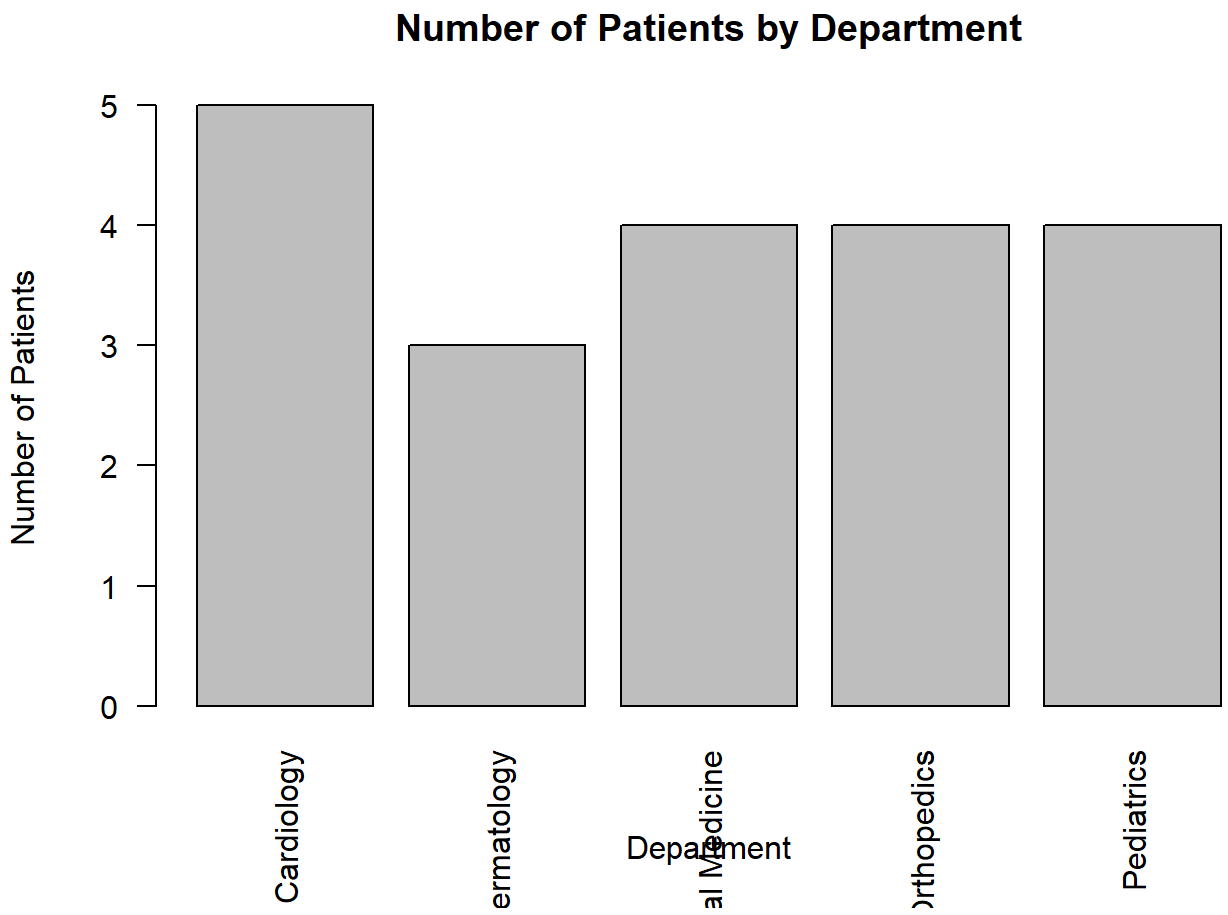
```
## [1] 92000
```

4. Patients by Department

```
department_count <- table(patient_data$Department)
department_count
```

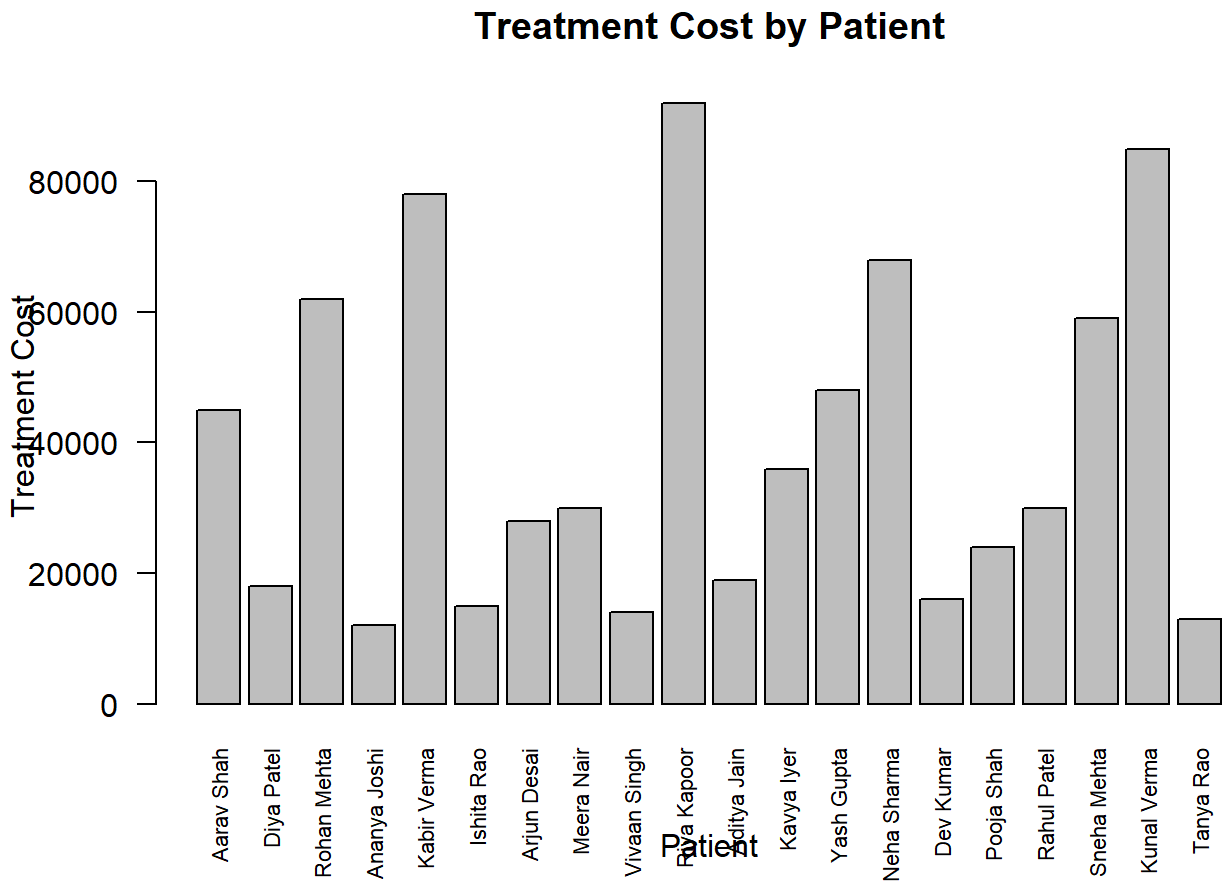
```
##
##       Cardiology      Dermatology General Medicine      Orthopedics
##                5                3                4                4
##       Pediatrics
##                4
```

```
barplot(department_count, main="Number of Patients by Department",
        xlab="Department", ylab="Number of Patients", las=2)
```



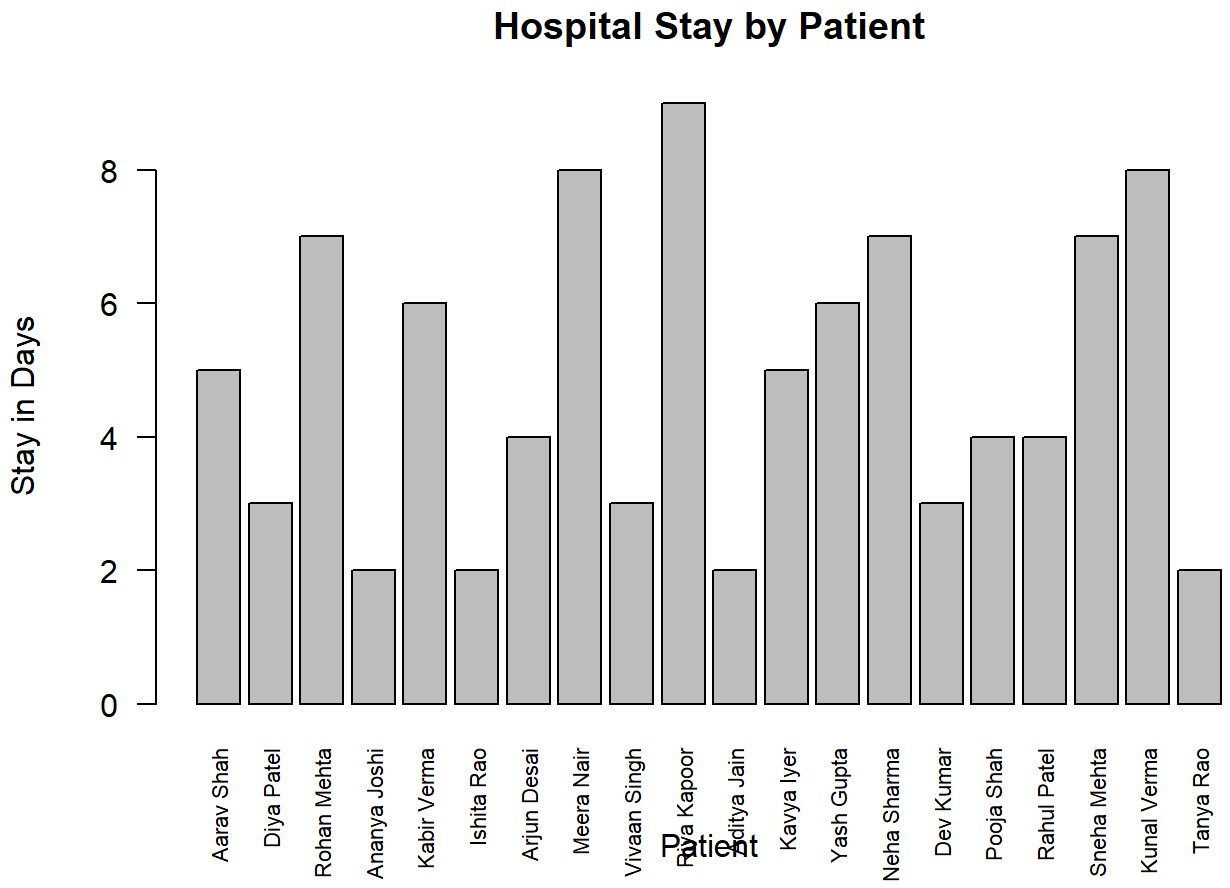
5. Treatment Cost by Patient

```
barplot(patient_data$Treatment_Cost, names.arg=patient_data$Patient_Name,
        main="Treatment Cost by Patient", xlab="Patient",
        ylab="Treatment Cost", las=2, cex.names=0.7)
```



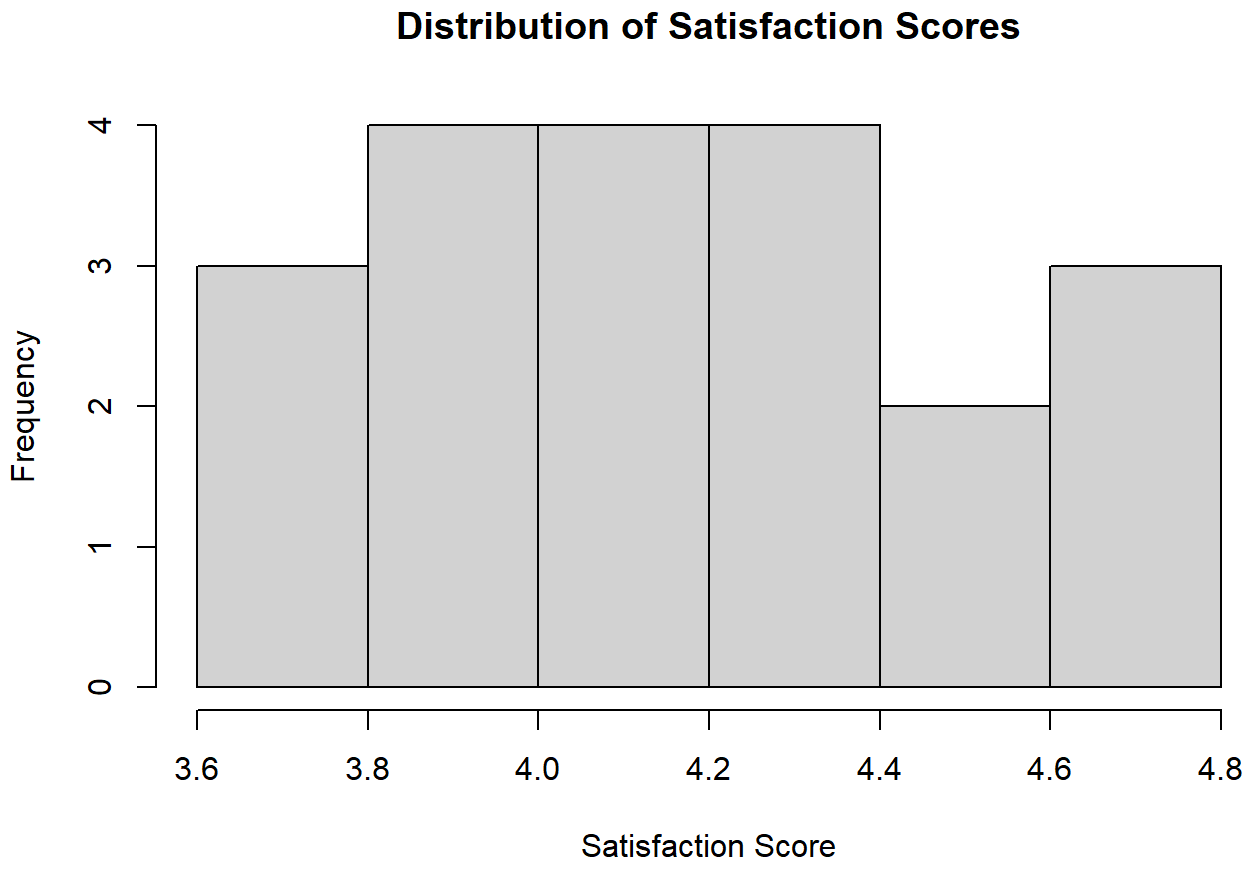
6. Hospital Stay by Patient

```
barplot(patient_data$Stay_Days, names.arg=patient_data$Patient_Name,
        main="Hospital Stay by Patient", xlab="Patient",
        ylab="Stay in Days", las=2, cex.names=0.7)
```



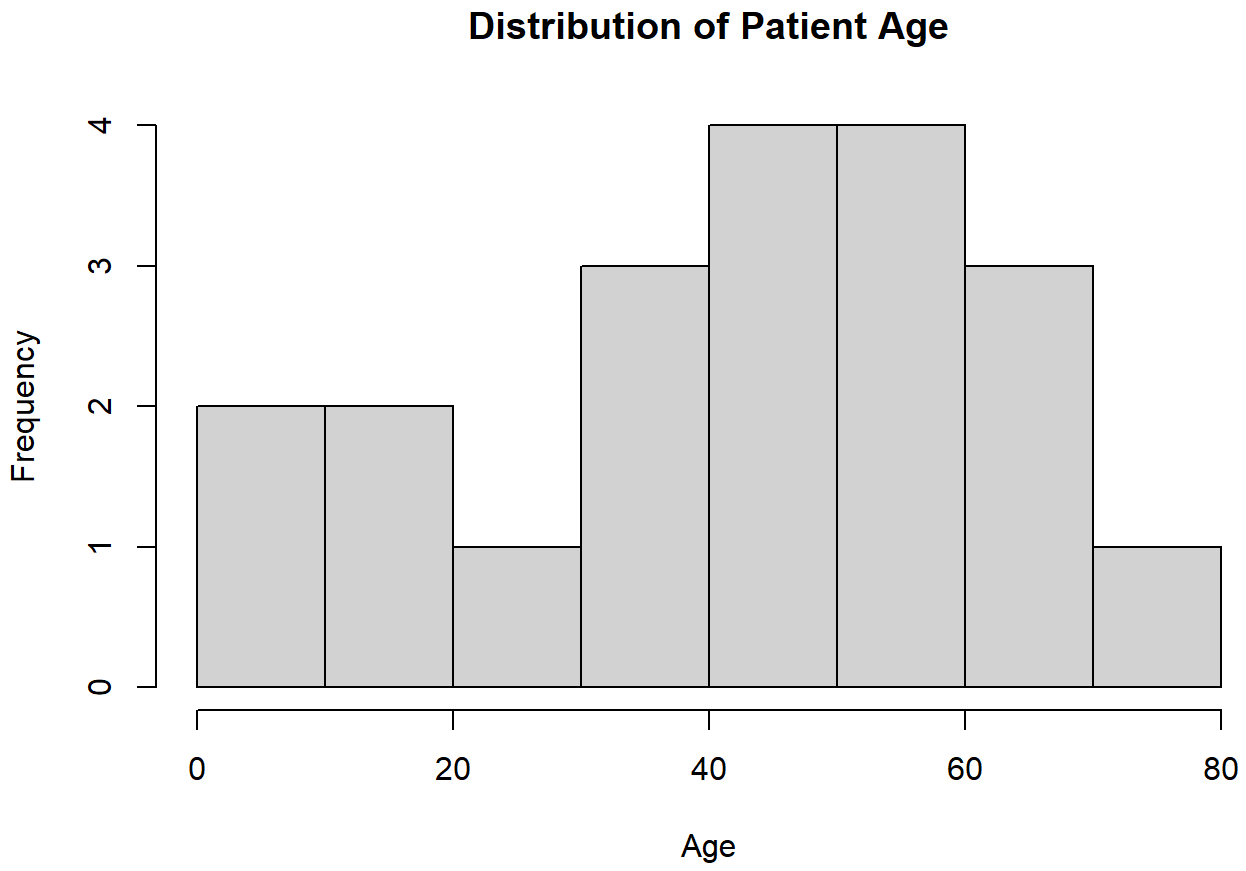
7. Satisfaction Score Distribution

```
hist(patient_data$Satisfaction_Score, main="Distribution of Satisfaction Scores",
     xlab="Satisfaction Score", ylab="Frequency")
```



8. Age Distribution

```
hist(patient_data$Age, main="Distribution of Patient Age",
     xlab="Age", ylab="Frequency")
```



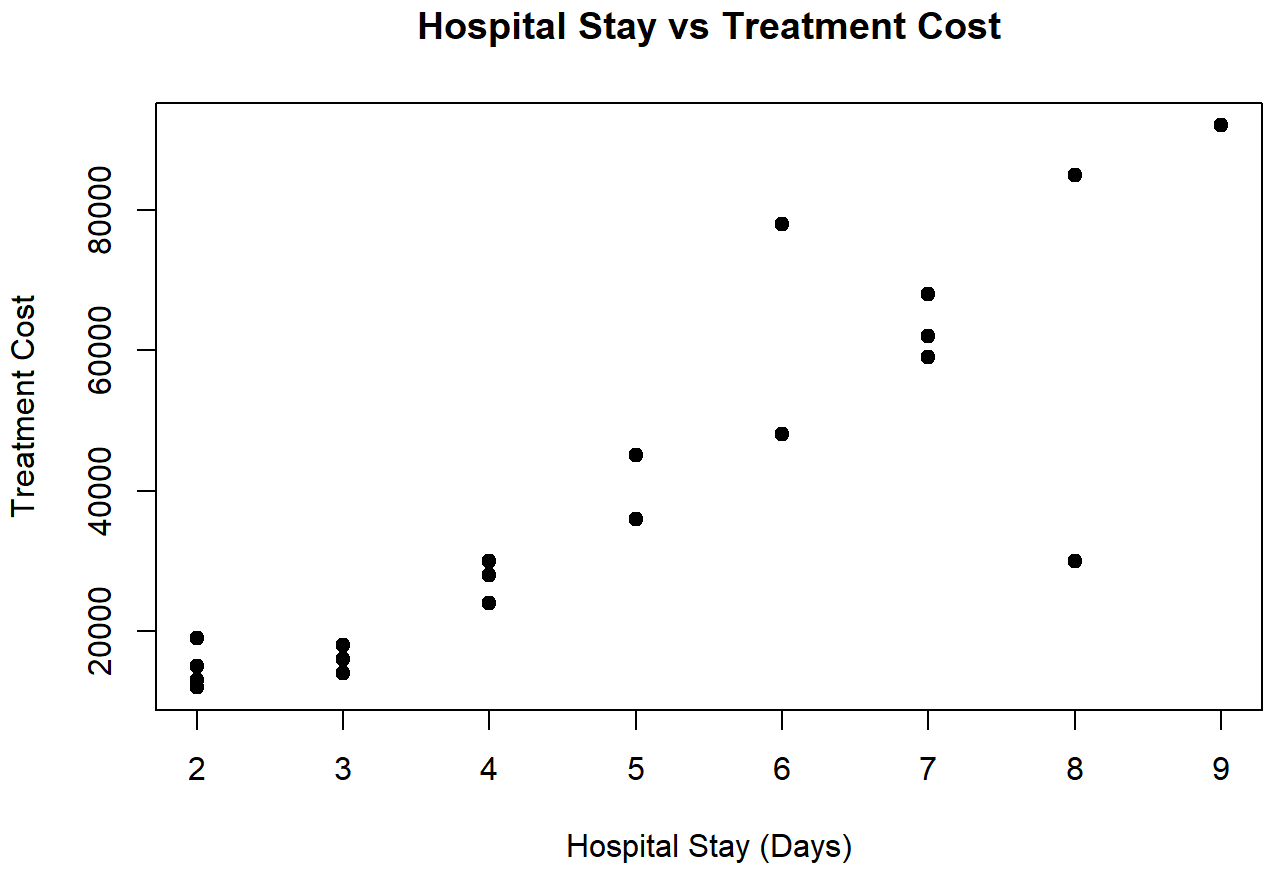
9. Correlation Analysis

Stay Days and Treatment Cost

```
cor(patient_data$Stay_Days, patient_data$Treatment_Cost)
```

```
## [1] 0.8726266
```

```
plot(patient_data$Stay_Days, patient_data$Treatment_Cost,
     main="Hospital Stay vs Treatment Cost",
     xlab="Hospital Stay (Days)", ylab="Treatment Cost", pch=19)
```

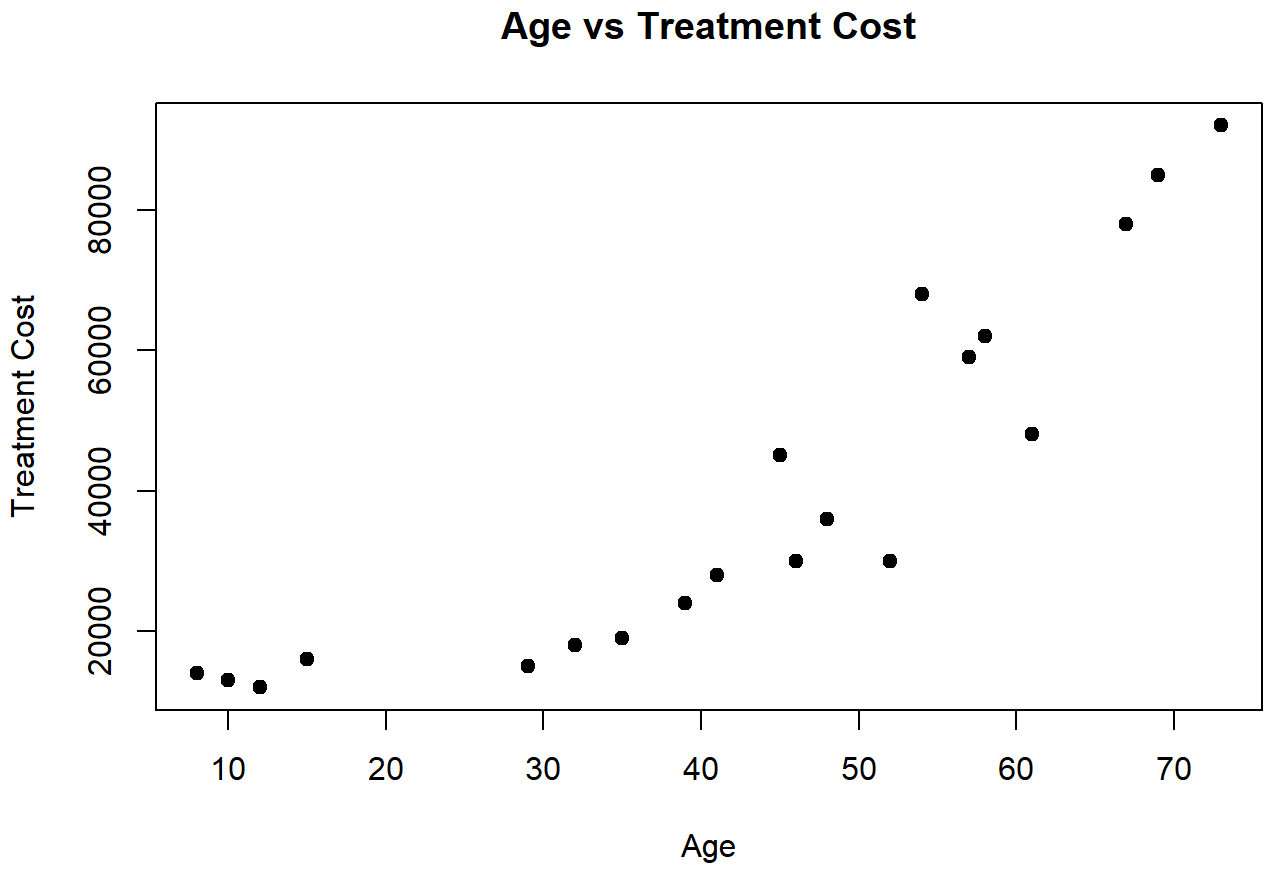


Age and Treatment Cost

```
cor(patient_data$Age, patient_data$Treatment_Cost)
```

```
## [1] 0.8864428
```

```
plot(patient_data$Age, patient_data$Treatment_Cost,
     main="Age vs Treatment Cost",
     xlab="Age", ylab="Treatment Cost", pch=19)
```



10. Findings

- Patient data was analyzed using age, hospital stay, treatment cost, and satisfaction score.
- Patients were compared across departments.
- Treatment costs and hospital stays were represented using bar plots.
- Age and satisfaction score distributions were examined using histograms.
- Correlation analysis was used to examine relationships between hospital stay, age, and treatment cost.

11. Conclusion

The Hospital Patient Analysis project demonstrates how R can be used to import, clean, analyze, and visualize healthcare-related data using statistical calculations, graphs, and correlation analysis.